Qualification Pack

Gas Tungsten Arc Welding (GTAW)

QP Code: ISC/Q0911

Version: 1.0

NSQF Level: 4

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Contents

ISC/Q0911: Gas Tungsten Arc Welding (GTAW)

Brief Job Description

This job is all about performing manual TIG (GTAW) welding for a range of standard welding job requirements. This is for a skilled welder who can weld different materials (carbon steel, aluminium, nickel, titanium, copper and stainless steel) in various positions and prepare various joints including corner, butt, fillet and tee. Set-up and prepare for operations interpreting the right information from the WPS.

Personal Attributes

The candidate should possess basic communication, numerical and computational abilities. Openness to learning, ability to plan and organize own work and identify and solve problems in the course of working. Understanding the need to take initiative and manage self and work to improve efficiency and effectiveness.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. ISC/N0008: Use basic health and safety practices at the workplace
2. ISC/N0009: Work effectively with others
3. ISC/N0910: Manually cut metal and metal alloys using oxy-fuel gases
4. ISC/N0911: Perform Tungsten Inert Gas (TIG) Welding also known as Gas Tungsten Arc Welding (GTAW)

Qualification Pack (QP) Parameters

Sector	Iron and Steel
Sub-Sector	Steel, Sponge Iron, Ferro Alloys, Re-Rollers, Refractory
Occupation	Mechanical Maintenance
Country	India
NSQF Level	4
Credits	NA
Aligned to NCO/ISCO/ISIC Code	NCO-2004/NIL

Minimum Educational Qualification & Experience	10th Class with 1-2 Years of experience In similar function OR 10th Class with 3-5 Years of experience In lieu of minimum qualification the incumbent should have 4-5 years of relevant work experience
Minimum Level of Education for Training in School	10th Class
Pre-Requisite License or Training	Welding Processes/fitment and precisions along with classification & Coding of welding electrodes Selection of Tungsten Rod composition, dia, gas requirement and purging Basic Welding Metallurgy and Weldability of metals Ferrous & Non-Ferrous Weld Defects/distortion - their stress, control, cause & remedies Welding Consumables and Control of welding parameters based on welding material
Minimum Job Entry Age	18 Years
Last Reviewed On	30/12/2014
Next Review Date	31/03/2022
NSQC Approval Date	18/06/2015
Version	1.0
Reference code on NQR	2015/IS/IISSSC/00296
NQR Version	1.0

ISC/N0008: Use basic health and safety practices at the workplace

Description

This OS unit is about knowledge and practices relating to health, safety and security that candidates need to use in the workplace. It covers responsibilities towards self, others, assets and the environment. It includes understanding of risks and hazards in the workplace, along with common techniques to minimize risk, deal with accidents, emergencies, etc

Elements and Performance Criteria

Health and safety

To be competent, the user/individual on the job must be able to:

PC1. Use protective clothing/equipment for specific tasks and work conditionsProtective clothing includes: Leather or asbestos gloves Flame proof aprons Flame proof overalls buttoned to neck Cuff less (without folds) trousers Reinforced footwear Helmets/hard hats Cap and shoulder covers Ear defenders/plugs Safety boots Knee pads Particle masks Glasses/gloves/visorsEquipment includes: Hand shields Machine guards Residual current devices Shields Dust sheets Respirator

Health and safetyprocedures

To be competent, the user/individual on the job must be able to:

PC2. State the name and location of people responsible for health and safety in the workplaceVarious areas are listed below: On chemical containers Equipment Packages Inside buildings Open areas and public spaces, etc.

PC3. State the names and location of documents that refer to health and safety in the workplace

PC4. Identify job-site hazardous work and state possible causes of risk or accident in the workplaceHazards include: Working with electrical and thermal tools and equipment Sharp edged and heavy tools Heated metals Oxyfuel and gas cylinders Welding radiation Surfaces: sharp, slippery, uneven, chipped, broken, etc. Substances: chemicals, gas, oxy-fuel, fumes, dust, etc. Physical: working at heights, large and heavy objects and machines, sharpand piercing objects, tolls and machines, intense light, load noise,obstructions in corridors, by doors, blind turns, noise, over stackedshelves and packages, etc. Electrical: power supply and points, loose and naked cables and wires,electrical machines and appliances, etc.

PC5. Carry out safe working practices while dealing with hazards to ensure the safety of self and others state methods of accident prevention in the work environment of the job roleSafe working practices include: Using protective clothing and equipment Putting up and reading safety signs Handle tools in the correct manner and store and maintain them properly Keep work area clear of clutter, spillage and unsafe object lying casually While working with electricity take all electrical precautions like insulatedclothing, adequate equipment insulation, use of control equipment, drywork area, switch off the power supply when not required, etc. Safe lifting and carrying practices Use equipment that is working properly and is well maintained Take due measures for safety while working in confined places, trenchesor at heights, etc. Including safety harness, fall arrestors, etc.Methods are: Training in health and safety procedures Using health and safety procedures Use of equipment and working practices (such as safe carryingprocedures) Safety notices, advice Instruction from colleagues and supervisors

PC6. State location of general health and safety equipment in the workplace

PC7. Inspect for faults, set up and safely use steps and ladders in general useFaults : Corrosion of metal components Deterioration Splits and cracks timber components Imbalance Loose rungs Nuts or bolts, etc.Set up: Firm/level base Clip/lash down Leaning at the correct angle, etc.

PC8. Work safely in and around trenches, elevated places and confined areas

PC9. Lift heavy objects safely using correct procedures

PC10. Apply good housekeeping practices at all times. Good housekeeping practices: Clean/tidy work areas Removal/disposal of waste products Protect surfaces

PC11. Identify common hazard signs displayed in various areas

PC12. Retrieve and/or point out documents that refer to health and safety in the workplace

Fire safety procedures

To be competent, the user/individual on the job must be able to:

PC13. Use the various appropriate fire extinguishers on different types of firescorrectly.Fire extinguishers: Sand Water Foam Co2 Dry powderFires: Class A: Ordinary solid combustibles, e.g. wood, paper, cloth, plastic,charcoal etc. Class B: Flammable liquids and gases, e.g. gasoline, propane, diesel fuel,tar, cooking oil and similar substances Class C: Electrical equipment e.g. appliances, wiring, breaker panels etc.(these categories of fires become Class A, B, and D fires when theelectrical equipment that initiated the fire is no longer receivingelectricity) Class D: Combustible metals such as magnesium, titanium, and sodium(these fires burn at extremely high temperatures and require specialsuppression agents)Causes of fires: Heating of metal Spontaneous ignition Sparking, Electrical heating Loose fires (e.g. Smoking, welding, etc.) Chemical fires, etc.

PC14. Demonstrate rescue techniques applied during fire hazard

PC15. Demonstrate good housekeeping in order to prevent fire hazards

PC16. Demonstrate the correct use of a fire extinguisher

Emergencies, rescue and first-aid procedures

To be competent, the user/individual on the job must be able to:

PC17. Demonstrate how to free a person from electrocution

PC18. Administer appropriate first aid to victims as required e.g. in case of bleeding, burns, choking, electric shock, poisoning etc.

PC19. Demonstrate basic techniques of bandaging

PC20. Respond promptly and appropriately to an accident situation or medical emergency in real or simulated environments. few General health and safety equipment are mentioned below : Fire extinguishers First aid equipment Safety instruments and clothing Safety installations, e.g. Fire exits, exhaust fans etc.

PC21. Perform and organize loss minimization or rescue activity during an accident in real or simulated environments

PC22. Administer first aid to victims in case of a heart attack or cardiac arrest due to electric shock, before the arrival of emergency services in real or simulated cases

PC23. Demonstrate the artificial respiration and the CPR Process

PC24. Participate in emergency procedures. Emergency procedures are: Raising alarm Safe/efficient evacuation Correct means of escape Correct assembly point Roll call Correct return to work

PC25. Complete a written accident/incident report or dictate a report to another person, and send report to person responsible Incident Report should capture: Name Date/time of incident Date/time of report, Location Environment conditions Persons involved Sequence of events Injuries sustained Damage sustained Actions taken Witnesses Supervisor/manager notifiedDocuments: Fire notices Accident reports Safety instructions for equipment and procedures Company notices and documents Legal documents (e.g. Government notices)Job titles:ISC/N0008: Use basic health and safety practices at the workplace Health and safety officer First aid officer Fire officer

PC26. Demonstrate correct method to move injured people and others during an emergency

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. State the names (and job titles if applicable), and describe where to find, all the people responsible for health and safety in a workplace

KU2. State the names and location of documents that refer to health and safety in the workplace

KU3. Meaning of hazards and risks

KU4. Health and safety hazards commonly present in the work environment and related precautions

KU5. Possible causes of risk, hazard or accident in the workplace and why risk and/or accidents are possible

KU6. Activities and causes of risk and accident

KU7. Methods of accident prevention

KU8. Safe working practices when working with tools and machines

KU9. Safe working practices while working at various hazardous sites

KU10. Where to find all the general health and safety equipment in the workplace

KU11. Various dangers associated with the use of electrical equipment

KU12. Preventative and remedial actions to be taken in the case of exposure to toxic materials. Exposure: ingested, contact with skin, inhaled Preventative action: ventilation, masks, protective clothing/equipment Remedial action: immediate first aid, report to supervisor Materials: solvents, flux, lead

KU13. Importance of using protective clothing/equipment while working

KU14. Precautionary activities to prevent the fire accidentActivities and causes: Physical actions Reading Listening to and giving instructions Inattention Sickness and incapacity (e.g. Drunkenness) Health hazards (e.g. Untreated injuries and contagious illness)

KU15. Various causes of fire

KU16. Techniques of using the different fire extinguishers

KU17. Different methods of extinguishing fire

KU18. Rescue techniques applied during a fire hazard

KU19. Various types of safety signs and what they mean

KU20. Appropriate basic first aid treatment relevant to the condition e.g. Shock, electrical shock, bleeding, breaks to bones, minor burns, resuscitation, poisoning, eye injuries

KU21. Content of written accident report

KU22. Potential injuries and ill health associated with incorrect manual handing
KU23. Safe lifting and carrying practices
KU24. Personal safety, health and dignity issues relating to the movement of a person by others
KU25. Potential impact to a person who is moved incorrectly

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Read and comprehend basic content to read labels, charts, signages
GS2. Read and comprehend basic English to read manuals of operations
GS3. Read and write an accident/incident report in local language or English
GS4. Question co-workers appropriately in order to clarify instructions and other issues
GS5. Give clear instructions to co-workers, subordinates others
GS6. Make appropriate decisions pertaining to the concerned area of work with respect to intended work objective, span of authority, responsibility, laid down procedure and guidelines
GS7. Plan and organize their own work schedule, work area, tools, equipment and materials to maintain decorum and for improved productivity
GS8. Remain congenial while discussing and debating issues with co-workers
GS9. Follow appropriate protocols for communication based on situation, hierarchy, organizational culture and practice
GS10. Ask for, provide and receive required assistance where possible to ensure achievement of work related objectives
GS11. Thank co-workers for any assistance received
GS12. Offer appropriate respect based on mutuality and respect for fellow workmanship and authority
GS13. Think through the problem, evaluate the possible solution(s) and suggest an optimum /best possible solution(s)
GS14. Identify immediate or temporary solutions to resolve delays
GS15. Identify sources of support that can be availed of for problem solving for various kind of problems
GS16. Report problems that you cannot resolve to appropriate authority
GS17. Identify cause and effect relations in their area of work
GS18. Use cause and effect relations to anticipate potential problems and their solution

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Health and safety	**5**	**5**	**-**	**-**
PC1. Use protective clothing/equipment for specific tasks and work conditionsProtective clothing includes: Leather or asbestos gloves Flame proof aprons Flame proof overalls buttoned to neck Cuff less (without folds) trousers Reinforced footwear Helmets/hard hats Cap and shoulder covers Ear defenders/plugs Safety boots Knee pads Particle masks Glasses/gloves/visorsEquipment includes: Hand shields Machine guards Residual current devices Shields Dust sheets Respirator	5	5	-	-
Health and safetyprocedures	**15**	**40**	**-**	**-**
PC2. State the name and location of people responsible for health and safety in the workplaceVarious areas are listed below: On chemical containers Equipment Packages Inside buildings Open areas and public spaces, etc.	-	4	-	-
PC3. State the names and location of documents that refer to health and safety in the workplace	-	1	-	-
PC4. Identify job-site hazardous work and state possible causes of risk or accident in the workplaceHazards include: Working with electrical and thermal tools and equipment Sharp edged and heavy tools Heated metals Oxyfuel and gas cylinders Welding radiation Surfaces: sharp, slippery, uneven, chipped, broken, etc. Substances: chemicals, gas, oxy-fuel, fumes, dust, etc. Physical: working at heights, large and heavy objects and machines, sharpand piercing objects, tolls and machines, intense light, load noise,obstructions in corridors, by doors, blind turns, noise, over stackedshelves and packages, etc. Electrical: power supply and points, loose and naked cables and wires,electrical machines and appliances, etc.	5	5	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC5. Carry out safe working practices while dealing with hazards to ensure the safety of self and others state methods of accident prevention in the work environment of the job roleSafe working practices include: Using protective clothing and equipment Putting up and reading safety signs Handle tools in the correct manner and store and maintain them properly Keep work area clear of clutter, spillage and unsafe object lying casually While working with electricity take all electrical precautions like insulatedclothing, adequate equipment insulation, use of control equipment, drywork area, switch off the power supply when not required, etc. Safe lifting and carrying practices Use equipment that is working properly and is well maintained Take due measures for safety while working in confined places, trenchesor at heights, etc. Including safety harness, fall arrestors, etc.Methods are: Training in health and safety procedures Using health and safety procedures Use of equipment and working practices (such as safe carryingprocedures) Safety notices, advice Instruction from colleagues and supervisors	5	5	-	-
PC6. State location of general health and safety equipment in the workplace	-	5	-	-
PC7. Inspect for faults, set up and safely use steps and ladders in general useFaults : Corrosion of metal components Deterioration Splits and cracks timber components Imbalance Loose rungs Nuts or bolts, etc.Set up: Firm/level base Clip/lash down Leaning at the correct angle, etc.	-	5	-	-
PC8. Work safely in and around trenches, elevated places and confined areas	-	4	-	-
PC9. Lift heavy objects safely using correct procedures	-	4	-	-
PC10. Apply good housekeeping practices at all times. Good housekeeping practices: Clean/tidy work areas Removal/disposal of waste products Protect surfaces	-	1	-	-
PC11. Identify common hazard signs displayed in various areas	5	1	-	-
PC12. Retrieve and/or point out documents that refer to health and safety in the workplace	-	5	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Fire safety procedures	**10**	**15**	**-**	**-**
PC13. Use the various appropriate fire extinguishers on different types of firescorrectly.Fire extinguishers: Sand Water Foam Co2 Dry powderFires: Class A: Ordinary solid combustibles, e.g. wood, paper, cloth, plastic,charcoal etc. Class B: Flammable liquids and gases, e.g. gasoline, propane, diesel fuel,tar, cooking oil and similar substances Class C: Electrical equipment e.g. appliances, wiring, breaker panels etc.(these categories of fires become Class A, B, and D fires when theelectrical equipment that initiated the fire is no longer receivingelectricity) Class D: Combustible metals such as magnesium, titanium, and sodium(these fires burn at extremely high temperatures and require specialsuppression agents)Causes of fires: Heating of metal Spontaneous ignition Sparking, Electrical heating Loose fires (e.g. Smoking, welding, etc.) Chemical fires, etc.	5	5	-	-
PC14. Demonstrate rescue techniques applied during fire hazard	5	5	-	-
PC15. Demonstrate good housekeeping in order to prevent fire hazards	-	1	-	-
PC16. Demonstrate the correct use of a fire extinguisher	-	4	-	-
Emergencies, rescue and first-aid procedures	**15**	**45**	**-**	**-**
PC17. Demonstrate how to free a person from electrocution	-	5	-	-
PC18. Administer appropriate first aid to victims as required e.g. in case of bleeding, burns, choking, electric shock, poisoning etc.	5	5	-	-
PC19. Demonstrate basic techniques of bandaging	-	5	-	-
PC20. Respond promptly and appropriately to an accident situation or medical emergency in real or simulated environments. few General health and safety equipment are mentioned below : Fire extinguishers First aid equipment Safety instruments and clothing Safety installations, e.g. Fire exits, exhaust fans etc.	5	5	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC21. Perform and organize loss minimization or rescue activity during an accident in real or simulated environments	-	5	-	-
PC22. Administer first aid to victims in case of a heart attack or cardiac arrest due to electric shock, before the arrival of emergency services in real or simulated cases	-	5	-	-
PC23. Demonstrate the artificial respiration and the CPR Process	-	5	-	-
PC24. Participate in emergency procedures. Emergency procedures are: Raising alarm Safe/efficient evacuation Correct means of escape Correct assembly point Roll call Correct return to work	-	4	-	-
PC25. Complete a written accident/incident report or dictate a report to another person, and send report to person responsible Incident Report should capture: Name Date/time of incident Date/time of report, Location Environment conditions Persons involved Sequence of events Injuries sustained Damage sustained Actions taken Witnesses Supervisor/manager notifiedDocuments: Fire notices Accident reports Safety instructions for equipment and procedures Company notices and documents Legal documents (e.g. Government notices)Job titles:ISC/N0008: Use basic health and safety practices at the workplace Health and safety officer First aid officer Fire officer	5	5	-	-
PC26. Demonstrate correct method to move injured people and others during an emergency	-	1	-	-
NOS Total	**45**	**105**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	ISC/N0008
NOS Name	Use basic health and safety practices at the workplace
Sector	Iron and Steel
Sub-Sector	Steel, Sponge Iron, Ferro Alloys, Re-Rollers, Refractory
Occupation	Mechanical Maintenance
NSQF Level	4
Credits	TBD
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

ISC/N0009: Work effectively with others

Description

This unit covers basic etiquette and competencies that a candidate is required to possess and demonstrate in their behaviour and interactions with others at the workplace

Elements and Performance Criteria

Ensure appropriate communication with superiors, peers and others as applicable at work place

To be competent, the user/individual on the job must be able to:

PC1. Accurately receive information and instructions from the supervisor and fellow workers, getting clarification where required

PC2. Accurately pass on information to authorized persons who require it and within agreed timescale and confirm its receipt

PC3. Provide information to others clearly, at a pace and in a manner that helps them to understand

Demonstrate appropriate behaviour and etiquette at work place

To be competent, the user/individual on the job must be able to:

PC4. Display helpful behaviour by assisting others in performing tasks in a positive manner, where required and possible

PC5. Consult with and assist others to maximize effectiveness and efficiency incarrying out tasks

PC6. Display appropriate communication etiquette while working

PC7. Display active listening skills while interacting with others at work

PC8. Use appropriate tone, pitch and language to convey politeness, assertiveness, care and professionalism

PC9. Demonstrate responsible and disciplined behaviours at the workplace

PC10. Escalate grievances and problems to

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Legislation, standards, policies, and procedures followed in the company relevant to own employment and performance conditions

KU2. Reporting structure, inter-dependent functions, lines and procedures in the work area

KU3. Relevant people and their responsibilities within the work area

KU4. Escalation matrix and procedures for reporting work and employment related issues

KU5. Various categories of people that one is required to communicate and co ordinate with in the organization

KU6. Importance of effective communication in the workplace

KU7. Importance of teamwork in organizational and individual success

KU8. Various components of effective communication

KU9. Key elements of active listening
KU10. Value and importance of active listening and assertive communication
KU11. Barriers to effective communication
KU12. Importance of tone and pitch in effective communication
KU13. Importance of avoiding casual expletives and unpleasant terms while communicating professional circles
KU14. How poor communication practices can disturb people, environment and cause problems for the employee, the employer and the customer
KU15. Importance of ethics for professional success
KU16. Importance of discipline for professional success
KU17. What constitutes disciplined behaviour for a working professional
KU18. Common reasons for interpersonal conflict
KU19. Importance of developing effective working relationships for professional success
KU20. Expressing and addressing grievances appropriately and effectively
KU21. Importance and ways of managing interpersonal conflict effectively

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Read and comprehend basic content to read labels, charts, signages
GS2. Read and comprehend basic English to read manuals of operations
GS3. Read and write an accident/incident report in local language or English
GS4. Question co-workers appropriately in order to clarify instructions and other issues
GS5. Provide clear instructions to co-workers, subordinates others
GS6. Make appropriate decisions pertaining to the concerned area of work with respect to intended work objective, span of authority, responsibility, laid down
GS7. Plan and organize their own work schedule, work area, tools, equipment and materials to maintain decorum and for improved productivity
GS8. Remain congenial while discussing and debating issues with co-workers
GS9. Follow appropriate protocols for communication based on situation, hierarchy, organizational culture and practice
GS10. Ask for, provide and receive required assistance where possible to ensure achievement of work related objectives
GS11. Thank co-workers for any assistance received
GS12. Offer appropriate respect based on mutuality and respect for fellow workmanship and authority
GS13. Think through the problem, evaluate the possible solution(s) and suggest an optimum /best possible solution(s)
GS14. Identify immediate or temporary solutions to resolve delays
GS15. Identify sources of support that can be availed of for problem solving for various kind of problems

GS16. Identify sources of support that can be availed of for problem solving for various kind of problems

GS17. Report problems that you cannot resolve to appropriate authority

GS18. Identify cause and effect relations in their area of work

GS19. Use cause and effect relations to anticipate potential problems and their solution

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Ensure appropriate communication with superiors, peers and others as applicable at work place	**10**	**20**	**-**	**-**
PC1. Accurately receive information and instructions from the supervisor and fellow workers, getting clarification where required	5	5	-	-
PC2. Accurately pass on information to authorized persons who require it and within agreed timescale and confirm its receipt	5	5	-	-
PC3. Provide information to others clearly, at a pace and in a manner that helps them to understand	-	10	-	-
Demonstrate appropriate behaviour and etiquette at work place	**20**	**50**	**-**	**-**
PC4. Display helpful behaviour by assisting others in performing tasks in a positive manner, where required and possible	5	5	-	-
PC5. Consult with and assist others to maximize effectiveness and efficiency incarrying out tasks	5	5	-	-
PC6. Display appropriate communication etiquette while working	-	10	-	-
PC7. Display active listening skills while interacting with others at work	-	10	-	-
PC8. Use appropriate tone, pitch and language to convey politeness, assertiveness, care and professionalism	5	5	-	-
PC9. Demonstrate responsible and disciplined behaviours at the workplace	5	10	-	-
PC10. Escalate grievances and problems to	-	5	-	-
NOS Total	**30**	**70**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	ISC/N0009
NOS Name	Work effectively with others
Sector	Iron and Steel
Sub-Sector	Steel, Sponge Iron, Ferro Alloys, Re-Rollers, Refractory
Occupation	Mechanical Maintenance
NSQF Level	4
Credits	TBD
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

ISC/N0910: Manually cut metal and metal alloys using oxy-fuel gases

Description

This unit is about competencies required for manual cutting operations using oxy-fuel gas such as oxy-acetylene. The person would be able to independently carry out oxyfuel cutting operations for as per welding procedure specification (WPS). The candidate will be able to cut different materials (mild carbon steel, high tensile and special steels, other materials) in various positions.The candidate cuts metal and metal alloys to required shape and size by gas flame manually. Examines material to be cut and marks it according to instruction of specification. Mounts template and sets cutting equipment to specifications. Makes necessary connections and fits required size of nozzle or burner in welding torch. Releases and regulates flow of gas in nozzle or burner, ignites and adjusts flame. Guides flame by hand along cutting line at required speed and cuts metal to required size. May use oxyacetylene or any other appropriate gas flame. This involves settingup and preparing for operations interpreting the right information from the WPS, obtaining the right consumables and raw materials, etc.

Scope

This unit/task covers the following:

- Work Safely all the time
- Prepare for cutting operations
- Carry out cutting operations
- Carry out test for accuracy
- Dealing with contingencies

Elements and Performance Criteria

Work safely all the time

To be competent, the user/individual on the job must be able to:

PC1. work safely at all times, complying with health and safety legislation, regulations and other relevant guidelines

PC2. take necessary safety precautions for gas cutting operations including equipment, processes and checks

Prepare for cutting operations

To be competent, the user/individual on the job must be able to:

PC3. interpret cutting procedure data sheets specifications

PC4. check regulators, hoses and check that valves are securely connected and free from leaks and damage

PC5. check equipment is calibrated and approved for use

PC6. check/fit the correct gas nozzle to the torch

PC7. ensure preheat and oxygen holes on the tips are clean

PC8. check that a flashback arrestor is fitted

PC9. set appropriate gas pressures

PC10. use the correct procedure for lighting, adjusting and extinguishing the flame

PC11. adjust torch valve for type of flame such as neutral, carburizing and oxidizing

PC12. follow sequence of operations such as pre-heating material and initiating cut

PC13. mark out the locations for cutting accurately and as per requirement

PC14. use appropriate and safe procedures for handling and storing of gas cylinders.the safety precautions (general) are as mentioned below: general workshop safety fire prevention general hazards manual lifting overhead lifting surface conditions stability of surrounding structures, furniture, etc.the safety precautions (gas cutting) are as mentioned below: safety from trailing hoses safety from naked flames appropriate fume and gases extraction/control measures safety from explosive gas mixtures and oxygen enrichment safety from spatter and hot metal (distance, ppe, proper handling andplacement) protection from live and other electrical components, including insulation,proper earthing, proper loading, etc. adequate lighting appropriate personal protective equipment suitable aprons gloves safety boots correctly fitting overalls suitable eye shields/goggles protection of self and others from the effects of the flame safety measures for elevated and trench working gas cylinder safety right colour coded correctly labelled no leakage away from heat or ignition source never use hose other than that designed for the specified gas use ferrules or clamps designed for the hose (not ordinary wire or othersubstitute) to connect hoses to fittings upright position (fuel gas) physical care to avoid damage and falls, throws and bumps move on trolleys, cap closed and without regulators valves closed on empty cylindersemergencies (safety procedures): sustained backfire in a blowpipe close the oxygen valve of the blowpipe, followed by the fuel valve andthen close both cylinder valves investigate the cause and rectify the fault re-light the blowpipe only after it is completely cooled down flashback into the hose and equipment, or a hose fire or explosion, or afire at the gas regulator connections isolate the fuel gas and oxygen supplies by closing the cylinder valves onlywhen this can be done safely may attempt to control the fire by fire-fighting equipment only whenactivate the fire alarm and call for the fire services department as perorganizational procedures fires involving acetylene cylinders always best dealt with by firemen from the fire services departmenthowever, the following initial response may be appropriate: cool the cylinder by spraying with water only if it is safe to do so close the cylinder valve to control the fire only if it is safe to do so evacuate the building by activating the fire alarm or by any other means toavoid explosion never move an acetylene cylinder involved in a fire or which has been affected by heat from a nearby fire even if it seems cooleddown

PC15. prepare the work area for the cutting activities

PC16. obtain the appropriate tools and equipment for the oxy-fuel gas cutting operations, and check that they are in a safe and usable condition

PC17. check that the oxy-fuel gas cutting equipment is set up for the operations to be performedtypes of oxy-fuel cutting equipment are: hand-held oxy-fuel gas cutting equipment simple, portable, track-driven cutting equipment (electrical or mechanical) fixed bench gas cutting equipmentprinciples of oxy-fuel cutting used are: oxygen cutting for materials which readily get oxidized oxides have lower melting points than the metals widely used for ferrous materials oxygen cutting is not used for materials like aluminium, bronze, mild steels which resist oxidation cutting of high carbon steels and cast irons require special attention due to formation of heat affected zone (haz) where structural transformation occurs

PC18. adjust cylinder valves and adjust regulator for operating pressure to achieve specifications for required operations

PC19. where appropriate, mark out the components for the required operations, using appropriate tools and techniques

PC20. perform trial cut to check for cut defects. kinds of cutting operations are: down-hand straight cuts (freehand) making straight cuts (track guided) cutting regular shapes cutting irregular shapes making angled cuts cutting chamfers making radial cuts gouging/flushing bevelled edge weld preparations cutting out holes

Carry out cutting operations

To be competent, the user/individual on the job must be able to:

PC21. operate the oxy-fuel gas cutting equipment to produce items/cut shapes to thedimensions and profiles specified into various forms mentioned below: plate rolled section pipe/tube solid bars

PC22. use various types of oxy-fuel gas cutting methodsvarious components used are: colour coded cylinder oxygen colour coded cylinder acetylene/lpg cylinder valve flashback arrestor set of nozzles gas lighter nozzle cutting tips pressure regulator pressure gauge non-return valves colour coded flexible hose trolleys torches (rose-bud heating, cutting, others)

PC23. perform various cutting operations correctly

PC24. produce thermal cuts in various forms of material (metal of 3mm and above)

PC25. produce cut profiles for various type of materials as mentioned under: mild steel high tensile/special steel other appropriate metal

PC26. produce thermally-cut components which meet specified quality criteria leavethe work area in a safe and tidy condition on completion of the cutting activitiesquality criteria used are: dimensional accuracy is within the tolerances specified on the drawing/specification, or within +/- 2mm angled/radial cuts are within specification requirements cuts are clean and smooth and free from flutes no dragsquality parameters are: shape and length of the draglines smoothness of the sides sharpness of the top edges amount of slag adhering to the metal

PC27. recognize and correct burn-back and flashback

PC28. detect and correct defects in cut

Carry out test for accuracy

To be competent, the user/individual on the job must be able to:

PC29. check that the finished components meet the standard required

PC30. use appropriate methods and equipment to check the quality, and that all dimensional and geometrical aspects of the cut material are to the specification

PC31. identify various cutting defects

Dealing with Contingencies

To be competent, the user/individual on the job must be able to:

PC32. report any difficulties or problems that may arise with the cutting activities, and carry out any agreed actions

PC33. detect equipment malfunctions and deal with them appropriately

PC34. deal promptly and effectively with problems within their control, and seek help and guidance from the relevant people if they have problems that they cannot resolve

PC35. shut down and make safe the cutting equipment on completion of the cutting activities

PC36. in case of emergencies follow standard emergency procedures

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. job relevant legislation, standards, policies, and procedures followed in the company
KU2. key purpose of the organization
KU3. department structure and hierarchy protocols
KU4. work flow and own role in the workflow
KU5. dependencies and interdependencies in the workflow
KU6. support functions and types of support available for incumbents in this role
KU7. types of fire extinguishers and their suitable uses in case of gas cutting related fires
KU8. specific safety precautions to be taken when working with oxy-fuel gas cutting equipment in a fabrication environment
KU9. personal protective clothing and equipment (ppe) to be worn when working with gas cutting equipment
KU10. hazards associated with carrying out gas cutting activities and how they can be minimized
KU11. safe working practices and procedures for using thermal equipment
KU12. principles of oxy-fuel gas cutting
KU13. procedure for obtaining the required drawings, job instructions and other related specifications
KU14. how to use and extract information from engineering drawings and related specifications, work piece reference points and system of tolerances
KU15. various types of gas cutting equipment available
KU16. various components of the gas cutting equipment
KU17. construction of the heating and cutting torch
KU18. types of oxy-fuel gases such as acetylene, natural gas and propane
KU19. accessories that can be used with handheld gas cutting equipment to aid cutting operations (such as cutting guides, trammels, templates)
KU20. importance and correct procedure for marking before a cut (eg. allowances for post-cut operations, punch marks, etc.)
KU21. types of regulators such as low- and high-pressure, and single- and two-stage
KU22. how to identify the gases used in the cutting process, and the colour coding of gas cylinders
KU23. type and thickness of base metals related to nozzle type
KU24. preparations prior to cutting (including checking connections for leaks, setting gas pressures, setting up the material/work piece, and checking the cleanliness of materials used)
KU25. holding methods that are used to aid thermal cutting, and the equipment that can be used.lighting and cutting procedures are mentioned below: lighting the cutting torch adjusting gas controls to produce a neutral flame methods of starting the cut and controlling the cutting speed direction and angle of cut procedure for extinguishing the flame
KU26. correct procedure for lighting, cutting and extinguishing the flame
KU27. types of flames and their implication for cutting
KU28. importance of following the correct procedure for lighting, cutting and extinguishing a flame
KU29. problems that can occur with thermal cutting, and how they can be avoided(including causes of distortion during thermal cutting and methods of controllingdistortion).defects that can occur in the (oxy-fuel cutting) process are: distortion grooved, fluted or ragged cuts poor draglines rounded edges tightly adhering slag
KU30. effects of oil, grease, scale or dirt on the cutting process

KU31. quality parameters for gas cut materials
KU32. causes of cutting defects, how to recognize them, and methods of correction and prevention
KU33. importance of leaving the work area in a safe and clean condition on completion of activities
KU34. correct handling and storage of gas cylinders
KU35. emergency procedures for backfires, flashback and other fires
KU36. how to close down the cutting equipment safely and correctly
KU37. purging tools and their function

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and interpret information correctly from various job specification documents, manuals, health and safety instructions, memos, etc. applicable to the job in english and/or local language
GS2. fill up appropriate technical forms, process charts, activity logs as per organizational format in english and/or local language
GS3. convey and share technical information clearly using appropriate language
GS4. check and clarify task-related information
GS5. liaise with appropriate authorities using correct protocol
GS6. communicate with people in respectful form and manner in line with organizational protocol
GS7. undertake numerical operations, geometry and calculations/ formulae (including addition, subtraction, multiplication, division, fractions and decimals, percentages and proportions, simple ratios and averages)
GS8. use appropriate measuring techniques
GS9. use and convert imperial and metric systems of measurements
GS10. apply appropriate degree of accuracy to express numbers
GS11. use tolerance in terms of limits of size
GS12. check measurements, angles, orientation and slopes
GS13. types of reference lines such as tangent lines, datum lines, centre lines and work points
GS14. check square of material using corner-to-corner dimensions and triangulation (3-4-5) method
GS15. select and use tools and equipment such as measuring tapes, levels, squares, protractors and dividers
GS16. ability to check dimensions of components
GS17. calculate the value of angles in a triangle
GS18. participate in on-the-job and other learning, training and development interventions and assessments
GS19. clarify task related information with appropriate personnel or technical adviser
GS20. seek to improve and modify own work practices
GS21. maintain current knowledge of application standards, legislation, codes of practice and product/process developments
GS22. identify problems with work planning, procedures, output and behaviour and their implication

GS23. prioritize and plan for problem solving
GS24. communicate problems appropriately to others
GS25. identify sources of information and support for problem solving
GS26. seek appropriate assistance from other sources to resolve problems
GS27. identify effective resolution techniques
GS28. select and apply resolution techniques
GS29. seek evidence for problem resolution
GS30. plan, prioritize and sequence work operations as per job requirements
GS31. organize and analyze information relevant to work
GS32. basic concepts of shop-floor work productivity including waste reduction, efficient material usage and optimization of time
GS33. importance and impact of initiative and enterprise for achieving better results for self, others and organization
GS34. how to undertake and express new ideas and initiatives to others
GS35. modify work plan to overcome unforeseen difficulties or developments that occur as work progresses
GS36. participate in improvement procedures including process, quality and internal/external customer/supplier relationships
GS37. ones competencies can and should be applied in new and different situations and contexts to achieve more
GS38. importance of taking responsibility for own work outcomes
GS39. importance of adherence to work timings, dress code and other organizational policies
GS40. importance of following laid down rules, procedures, instructions and policies
GS41. importance of exercising restraint while expressing dissent and during conflict situations
GS42. how to avoid and manage distractions to be disciplined at work
GS43. importance of time management for achieving better results
GS44. work in a team in order to achieve better results
GS45. identify and clarify work roles within a team
GS46. communicate and cooperate with others in the team
GS47. seek assistance from fellow team members

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Work safely all the time	**10**	**10**	**-**	**-**
PC1. work safely at all times, complying with health and safety legislation, regulations and other relevant guidelines	5	5	-	-
PC2. take necessary safety precautions for gas cutting operations including equipment, processes and checks	5	5	-	-
Prepare for cutting operations	**25**	**105**	**-**	**-**
PC3. interpret cutting procedure data sheets specifications	5	5	-	-
PC4. check regulators, hoses and check that valves are securely connected and free from leaks and damage	-	5	-	-
PC5. check equipment is calibrated and approved for use	-	5	-	-
PC6. check/fit the correct gas nozzle to the torch	-	5	-	-
PC7. ensure preheat and oxygen holes on the tips are clean	5	5	-	-
PC8. check that a flashback arrestor is fitted	5	10	-	-
PC9. set appropriate gas pressures	-	5	-	-
PC10. use the correct procedure for lighting, adjusting and extinguishing the flame	-	5	-	-
PC11. adjust torch valve for type of flame such as neutral, carburizing and oxidizing	-	5	-	-
PC12. follow sequence of operations such as pre-heating material and initiating cut	5	5	-	-
PC13. mark out the locations for cutting accurately and as per requirement	-	5	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. use appropriate and safe procedures for handling and storing of gas cylinders.the safety precautions (general) are as mentioned below: general workshop safety fire prevention general hazards manual lifting overhead lifting surface conditions stability of surrounding structures, furniture, etc.the safety precautions (gas cutting) are as mentioned below: safety from trailing hoses safety from naked flames appropriate fume and gases extraction/control measures safety from explosive gas mixtures and oxygen enrichment safety from spatter and hot metal (distance, ppe, proper handling andplacement) protection from live and other electrical components, including insulation,proper earthing, proper loading, etc. adequate lighting appropriate personal protective equipment suitable aprons gloves safety boots correctly fitting overalls suitable eye shields/goggles protection of self and others from the effects of the flame safety measures for elevated and trench working gas cylinder safety right colour coded correctly labelled no leakage away from heat or ignition source never use hose other than that designed for the specified gas use ferrules or clamps designed for the hose (not ordinary wire or othersubstitute) to connect hoses to fittings upright position (fuel gas) physical care to avoid damage and falls, throws and bumps move on trolleys, cap closed and without regulators valves closed on empty cylindersemergencies (safety procedures): sustained backfire in a blowpipe close the oxygen valve of the blowpipe, followed by the fuel valve andthen close both cylinder valves investigate the cause and rectify the fault re-light the blowpipe only after it is completely cooled down flashback into the hose and equipment, or a hose fire or explosion, or afire at the gas regulator connections isolate the fuel gas and oxygen supplies by closing the cylinder valves onlywhen this can be done safely may attempt to control the fire by fire-fighting equipment only whenactivate the fire alarm and call for the fire services department as perorganizational procedures fires involving acetylene cylinders always best dealt with by firemen from the fire services departmenthowever, the following initial response may be appropriate: cool the cylinder by spraying with water only if it is safe to do so close the cylinder valve to control the fire only if it is safe to do so evacuate the building by activating the fire alarm or by any other means toavoid explosion never move an acetylene cylinder involved in a fire or which has been affected by heat from a nearby fire even if it seems cooleddown	-	5	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC15. prepare the work area for the cutting activities	-	5	-	-
PC16. obtain the appropriate tools and equipment for the oxy-fuel gas cutting operations, and check that they are in a safe and usable condition	5	5	-	-
PC17. check that the oxy-fuel gas cutting equipment is set up for the operations to be performedtypes of oxy-fuel cutting equipment are: hand-held oxy-fuel gas cutting equipment simple, portable, track-driven cutting equipment (electrical or mechanical) fixed bench gas cutting equipmentprinciples of oxy-fuel cutting used are: oxygen cutting for materials which readily get oxidized oxides have lower melting points than the metals widely used for ferrous materials oxygen cutting is not used for materials like aluminium, bronze, mild steels which resist oxidation cutting of high carbon steels and cast irons require special attention due to formation of heat affected zone (haz) where structural transformation occurs	-	5	-	-
PC18. adjust cylinder valves and adjust regulator for operating pressure to achieve specifications for required operations	-	10	-	-
PC19. where appropriate, mark out the components for the required operations, using appropriate tools and techniques	-	10	-	-
PC20. perform trial cut to check for cut defects. kinds of cutting operations are: down-hand straight cuts (freehand) making straight cuts (track guided) cutting regular shapes cutting irregular shapes making angled cuts cutting chamfers making radial cuts gouging/flushing bevelled edge weld preparations cutting out holes	-	5	-	-
Carry out cutting operations	**20**	**45**	**-**	**-**
PC21. operate the oxy-fuel gas cutting equipment to produce items/cut shapes to thedimensions and profiles specified into various forms mentioned below: plate rolled section pipe/tube solid bars	5	5	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC22. use various types of oxy-fuel gas cutting methodsvarious components used are: colour coded cylinder oxygen colour coded cylinder acetylene/lpg cylinder valve flashback arrestor set of nozzles gas lighter nozzle cutting tips pressure regulator pressure gauge non-return valves colour coded flexible hose trolleys torches (rose-bud heating, cutting, others)	5	5	-	-
PC23. perform various cutting operations correctly	-	5	-	-
PC24. produce thermal cuts in various forms of material (metal of 3mm and above)	-	5	-	-
PC25. produce cut profiles for various type of materials as mentioned under: mild steel high tensile/special steel other appropriate metal	5	10	-	-
PC26. produce thermally-cut components which meet specified quality criteria leavethe work area in a safe and tidy condition on completion of the cutting activitiesquality criteria used are: dimensional accuracy is within the tolerances specified on the drawing/specification, or within +/- 2mm angled/radial cuts are within specification requirements cuts are clean and smooth and free from flutes no dragsquality parameters are: shape and length of the draglines smoothness of the sides sharpness of the top edges amount of slag adhering to the metal	-	5	-	-
PC27. recognize and correct burn-back and flashback	5	5	-	-
PC28. detect and correct defects in cut	-	5	-	-
Carry out test for accuracy	**10**	**30**	**-**	**-**
PC29. check that the finished components meet the standard required	5	10	-	-
PC30. use appropriate methods and equipment to check the quality, and that all dimensional and geometrical aspects of the cut material are to the specification	-	10	-	-
PC31. identify various cutting defects	5	10	-	-
Dealing with Contingencies	**20**	**25**	**-**	**-**
PC32. report any difficulties or problems that may arise with the cutting activities, and carry out any agreed actions	5	5	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC33. detect equipment malfunctions and deal with them appropriately	-	5	-	-
PC34. deal promptly and effectively with problems within their control, and seek help and guidance from the relevant people if they have problems that they cannot resolve	5	5	-	-
PC35. shut down and make safe the cutting equipment on completion of the cutting activities	5	5	-	-
PC36. in case of emergencies follow standard emergency procedures	5	5	-	-
NOS Total	**85**	**215**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	ISC/N0910
NOS Name	Manually cut metal and metal alloys using oxy-fuel gases
Sector	Iron and Steel
Sub-Sector	Steel, Sponge Iron, Ferro Alloys, Re-Rollers, Refractory
Occupation	Mechanical Maintenance
NSQF Level	4
Credits	TBD
Version	1.0
Last Reviewed Date	30/12/2014
Next Review Date	31/03/2022
NSQC Clearance Date	18/06/2015

ISC/N0911: Perform Tungsten Inert Gas (TIG) Welding also known as Gas Tungsten Arc Welding (GTAW)

Description

This unit covers the performing of manual TIG (GTAW) welding for a range of standard welding job requirements. This involves welding different materials (carbon steel, aluminium and stainless steel) in various positions. The welder can prepare various joints including corner, butt, fillet and tee. This involves setting-up and preparing for operations interpreting the right information from the WPS, obtaining the right consumables and raw materials, etc. The candidate will be expected to work with a minimum of supervision, taking personal responsibility for own actions, quality and accuracy of the work. The breakdown servicing activity may be carried out as a team effort, but the candidate would be responsible for the overall completion of the installation activities as per specifications.

Scope

This unit/task covers the following:

- Working Safely at all times
- Preparing for welding operations
- Carrying out welding operations
- Testing for quality
- Post welding techniques
- Dealing with contingencies

Elements and Performance Criteria

Working Safely at all times

To be competent, the user/individual on the job must be able to:

PC1. work safely at all times, complying with health and safety legislation, regulations and other relevant guidelinessafety precautions (general) are: general workshop safety fire prevention general hazards manual lifting overhead lifting / mechanised lifting surface conditions stability of surrounding structures, furniture, etc.safety precautions (tig welding) are: protection from live and other electrical components, including insulation, proper earthing, proper loading, etc. proper handling and placement of hot metal adequate lighting appropriate personal protective equipment suitable aprons welding gloves safety boots correctly fitting overalls suitable welding helmet protection of self and others from the effects of the welding arc fume extraction/control measures safety measures for elevated and trench working reduction in the local air concentration due to release of argon gas during welding in confined places

PC2. take necessary safety precautions for tig welding operations

PC3. adhere to procedures and system in place for health and safety, pper and other regulations

PC4. check all connections of machines, welding leads, gas connection arrangement, electrode holder

Preparing for welding operations

To be competent, the user/individual on the job must be able to:

PC5. interpret weld procedure data sheets specificationswelding concepts and mechanisms used are: rated output (duty cycle) measurement of electrical output and continuity relationship between wire feed speed control and welding current power source characteristics (volt/ampere graph, flat characteristic, constantvoltage output) types of current ac and dc and polarity ac welding (square wave forms and wave balancing) dc pulsed tig welding function of induction (principle, effect, fixed, stepped, variable control) return earth indirect control of welding current relay for electrical powerwelding techniques used are: fine adjustment of parameters (current and gas flow) selection of gas nozzle if required selection of the outer nozzle correct manipulation of the torch blending in stops/starts and tack welds starting techniques

PC6. select welding machines e.g. inverters, rectifiers and generators, according to the materials and task

PC7. select proper welding torch and electrode(w) that meet the job requirement and specification, select tungsten electrode by the colour of the tip according to base metal, and correct diameterselection and preparation of tungsten electrode are: types and classification of tungsten electrodes for different materials angle and technique of preparation of the tungsten electrode tips selection of the tungsten electrode diameter as per currenttorch components are: cables water cooled cables ceramic nozzle collet collet holder gas lens

PC8. obtain filler wire according to specifications

PC9. prepare for the tig welding process

PC10. prepare the materials and joint in readiness for weldingmaterial and joint preparation activities are: made rust free cleaned free from scaling, paint, oil/grease chemical cleaning made dry and free from moisture edges to be welded prepared as per job requirement (e.g. flat, square or bevelled) use various machines and techniques for the above (e.g. chamfering machine grinding and stripping, gas and plasma cutting, etc.) correctly positioned positioning: devices and techniques jigs and fixtures restraining devices such as clamps and weights/blocks setting up the joint in the correct position and alignment

PC11. select and fit the welding shielding gases for a range of given applicationsincluding back purgingshielding gases: shielding gases for gtaw applications for shielding gases/gas mixtures (argon, argon/helium mixtures,argon/hydrogen mixtures, nitrogen argon/nitrogen mixtures) gas pressure requirements flow rates for applications back purgingshielding gases equipment are: cylinders manifold systems regulators (fixed, single stage, two-stage) gas flow meters gas tubes and connectors use of solenoid valves economisers

PC12. plan the welding activities before they start them effectively and efficiently forachieving specifications as per wpsinterpreting the wps: welding process (iso codes for e.g. aws/asme) parent metal consumables pre welding joint preparation (cleaning, edge preparation, assembly, pre-heat) welding parameters welding positions (en iso 6947 pa, pb, pc, pd, pe, pf, pg; asme ix i-6 g/1-6 f) number and arrangement of runs to fully fill/weld joints electrode (w) filler wire electrical conditions required (type of current, alternating [a.c.] direct [d.c.], electrode polarity (negative), welding current ranges, electrode polarity (positive) methods of arc ignition (scratch, high frequency, lift start), carbon block shielding gas (type, flow rate, pre-weld gas flow, post-weld gas flow), techniques (including autogenous) control of heat input inter-pass/run cleaning/back gouging methods, post welding activities (wiring brushing, removal of excess weld metal where required), post-weld heat treatment (normalising, stress relief) where permissible and restrictions activities to be checked before start of welding are: correct set-up of the joint proper condition of electrical connections welding return and earthing arrangements operating parameters

PC13. connect torches and components

PC14. connect and adjust regulators and flow meters to cylinders

PC15. read, set and adjust current (amperage) as required

PC16. set pre-purge with shielding gas as required

PC17. prepare tungsten by sharpening or balling it to desired tip shape

PC18. set and verify gas flow rates

PC19. prepare and support the joint, using the appropriate methods

PC20. tack weld the joint at appropriate intervals, and check the joint for accuracy before final welding, wherever required

PC21. match feed and travel speed as required

Carrying out welding operations

To be competent, the user/individual on the job must be able to:

PC22. perform tig welding operations to meet welding procedure specification requirementsbasic principles of tig welding are: the arc burns between a non- consumable tungsten electrode and the work piece exclusively inert gases (argon, helium) are used as shielding gases and other gases and gas mixtures tig welding installation for most applications an electrode with a negative polarity is used for welding of aluminium, alternating current must be used for arc ignition a high-frequency high voltage is used

PC23. use correct technique for starting the arc (using hf (high frequency) unit, scratching the electrode on the job material, lifting the electrode immediately after touching the job material)

PC24. use correct angle of torch and filler wire, direction of weld and inclusion defect

PC25. weld the joint to the specified quality, dimensions and profile.

PC26. use manual welding and related equipment, to carry out tig welding processes.welding equipment are: rectifier (pulsing) inverter generator equipment for current regulation high frequency unit torches electrodes filler wires water cooling and circulation system for tig torch (water/air cooled torch) return clamps foot pedal ancillary equipment (tungsten tip grinder for tungsten electrode, wirebrushes, linishers, hammer, power saw, angle, pedestal and straightgrinders, chisel) other equipments such as holding, jig fixtures, measuring equipments etc.

PC27. use welding consumables appropriate to the material and application, to include ac current types and dc current types

PC28. produce joints of the required quality and of specified dimensional accuracy which achieve a weld quality equivalent to level b of iso 5817weld quality check standards are: required parameters for dimensional accuracy weld finishes are built up to the full section of the weld joins at stop/start positions merge smoothly weld surface is o free from cracks substantially free from porosity free from any pronounced hump or crater substantially free from shrinkage cavities substantially free from arcing or chipping marks fillet welds are o equal in leg length slightly convex in profile (where applicable and preferable) size of the fillet equivalent to the lower thickness of the material welded weld contour is of linear and of uniform profile smooth and free from excessive undulations regular and has an even ripple formation welds are adequately fused, and there is minimal undercut, overlap and surface inclusions tack welds are blended in to form part of the finished weld, without excessive hump corner joints have minimal burn through to the underside of the joint or, where appropriatetypes of joints are: fillet lap joints tee fillet joints corner joints butt joints square single vee double vee

PC29. roduce joints from various materials in different forms.materials used for welding are: carbon steel stainless steel (all grades) aluminium and aluminium alloys nickel and nickel alloys titanium copper and copper alloys forms on which welding done are: sheet (less than 1.5 mm) plate (8 mm) section pipe/tube other forms

Testing for quality

To be competent, the user/individual on the job must be able to:

PC30. use appropriate methods and equipment to check the quality, and that all dimensional and geometrical aspects of the weld are to the specification

PC31. check that the welded joint conforms to the specification, by checking various quality parameters using visual inspectionquality parameters are: shape and length of the draglines smoothness of the sides sharpness of the top edges amount of slag adhering to the metalquality parameters a to be checked are: dimensional accuracy alignment/squareness size and profile of weld visual defects ndt/dt tested defects types of visual inspections are: use of visual techniques lighting low powered magnification fillet weld gauges

PC32. identify various weld defects;types of weld defects are: lack of continuity of the weld uneven and irregular ripple formation incorrect weld size or profile undercutting overlap inclusions (tungsten) porosity internal cracks surface cracks lack of fusion lack of penetration gouges stray arc strikes sharp edgeswelding consumables used are: filler wires for different base materials shielding gasconsumables classification as per: sizes [diameters, lengths] strength and elongation of the weld metal impact properties of the weld metal chemical composition of the weld metal protection of bare wires

PC33. detect surface imperfections and deal with them appropriately

PC34. carry out dpt tests to assess fine defect open to the surface not detected by visual inspection (vt)

Post welding techniques

To be competent, the user/individual on the job must be able to:

PC35. prepare for non-destructive testing of the welds for a range of testsnon-destructive tests (ndt) are: visual inspection leak test dye penetrant (dpt) fluorescent penetrant (fpt) magnetic particle (mpt) radiographic (rt) ultrasonic (ut)

PC36. prepare for destructive tests on weld specimens for select testsdestructive tests (dt) are: nick break test bend tests (such as face, root or side, as appropriate) metallographic (micro structure, haz, etc.) mechanical (peel, tensile and shear, fatigue, impact tests) and hardness in different zones chemicalhandling specimens for tests: handling hot materials using chemicals for cleaning and etching using equipment to fracture welds

PC37. shut down and make safe the welding equipment and gases on completion of the welding activities, clean the area & store the accessories in designated place

Dealing with Contingencies

To be competent, the user/individual on the job must be able to:

PC38. detect equipment malfunctions and deal with them appropriately

PC39. deal promptly and effectively with problems within their control, and seek help and guidance from the relevant people if they have problems that they cannot resolve

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. legislation, standards, policies, and procedures followed in the company
KU2. purpose of the organization
KU3. structure and hierarchy protocols
KU4. flow and own role in the workflow
KU5. and interdependencies in the workflow
KU6. functions and types of support available for incumbents in this role
KU7. types of fire extinguishers and their suitable uses in case of welding related fires
KU8. effects of exposure to welding fume
KU9. of welding equipment available
KU10. of welding equipment
KU11. types of power source
KU12. to compare welding consumables for suitability for a range of given applications
KU13. consumables classification
KU14. working practices and procedures to be followed when preparing and using tig welding equipment
KU15. associated with tig welding and safety precautions to minimize risk
KU16. variants of the tig welding (eg. orbital welding, internal bore welding, ng-tig etc.)
KU17. protective equipment to be worn for the welding activities
KU18. handling and storage of gas cylinders
KU19. tig welding process
KU20. and thickness of base metals
KU21. types and polarity
KU22. of tungsten
KU23. selection and application of filler wires and welding electrodes
KU24. for using shielding gases, and the types and application of the various gases and gas mixtures
KU25. of shielding gas composition and purity on welding quality
KU26. impact and importance of gas pressures and flow rates in relationship to the type of material being welded
KU27. and post-flow purge and its importance
KU28. and application of back purging
KU29. of welded joints to be produced
KU30. used for the appropriate welding positionswelding positions are: flat (pa) ig/1f horizontal vertical (pb) 2f horizontal (pc) 2g vertical upwards (pf) 3f / 3g vertical downwards (pg) 3f / 3g plate to pipe (fixed) 5f pipe to pipe 5g pipe welding at inclined position 6g
KU31. of torches such as air cooled and water cooled
KU32. how to prepare the materials in readiness for the welding activity
KU33. how to set up and restrain the joint, and the tools and techniques to be used
KU34. appropriate tack welding size and spacing (in relationship to material thickness)
KU35. checks to be made prior to welding

KU36. techniques of operating the welding equipment to produce a range of joints in the various joint positions

KU37. effects of the electrical characteristics of the tig welding arcelectrical characteristics are: power source characteristics (volt/ampere graph, drooping characteristic,constant current output) effects of types of current and electrode polarity heat input/distribution electrode weld bead profile penetration methods of a.c. arc stabilisation (including: square wave) welding current features (pulse current, slope in, slope out) voltage (open circuit, arc)

KU38. how to control distortion (such as welding sequence; deposition technique)

KU39. problems that can occur with the welding activities

KU40. how to close down the welding equipment safely and correctly

KU41. how to prepare the welds for examination

KU42. how to check the welded joints for uniformity, alignment, position, weld size and profile

KU43. various procedures for visual examination of the welds for cracks / defects

KU44. non-destructive and destructive tests

KU45. methods of removing a test piece of weld from a suitable position in the joint

KU46. safe working practices and procedures to be adopted when preparing the welds for examination

KU47. importance of leaving the work area and equipment in a safe condition on completion of the welding activities

KU48. safe handling and recording of welded pieces

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and interpret information correctly from various job specification documents, manuals, health and safety instructions, memos, etc. applicable to the job in english and/or local language

GS2. fill up appropriate technical forms, process charts, activity logs as per organizational format in english and/or local language

GS3. convey and share technical information clearly using appropriate language

GS4. check and clarify task-related information

GS5. liaise with appropriate authorities using correct protocol

GS6. communicate with people in respectful form and manner in line with organizational protocol

GS7. undertake numerical operations, geometry and calculations/ formulae (including addition, subtraction, multiplication, division, fractions and decimals, percentages and proportions, simple ratios and averages)

GS8. use appropriate measuring techniques

GS9. use and convert imperial and metric systems of measurements

GS10. apply appropriate degree of accuracy to express numbers

GS11. use tolerance in terms of limits of size

GS12. check measurements, angles, orientation and slopes

GS13. types of reference lines such as tangent lines, datum lines, centre lines and work points

GS14. check square of material using corner-to-corner dimensions and triangulation (3-4-5) method
GS15. select and use tools and equipment such as measuring tapes, levels, squares, protractors and dividers
GS16. ability to check dimensions of components
GS17. calculate the value of angles in a triangle
GS18. participate in on-the-job and other learning, training and development interventions and assessments
GS19. clarify task related information with appropriate personnel or technical adviser
GS20. seek to improve and modify own work practices
GS21. maintain current knowledge of application standards, legislation, codes of practice and product/process developments
GS22. identify problems with work planning, procedures, output and behavior and their implications
GS23. prioritize and plan for problem solving
GS24. communicate problems appropriately to others
GS25. identify sources of information and support for problem solving
GS26. seek assistance and support from other sources to solve problems
GS27. identify effective resolution techniques
GS28. select and apply resolution techniques
GS29. seek evidence for problem resolution
GS30. plan, prioritize and sequence work operations as per job requirements
GS31. organize and analyze information relevant to work
GS32. basic concepts of shop-floor work productivity including waste reduction, efficient material usage and optimization of time
GS33. importance and impact of initiative and enterprise for achieving better results for self, others and organization
GS34. how to undertake and express new ideas and initiatives to others
GS35. modify work plan to overcome unforeseen difficulties or developments thatoccur as work progresses
GS36. participate in improvement procedures including process, quality and internal/external customer/supplier relationships
GS37. ones competencies can and should be applied in new and different situations and contexts to achieve more
GS38. importance of taking responsibility for own work outcomes
GS39. importance of adherence to work timings, dress code and other organizational policies
GS40. importance of following laid down rules, procedures, instructions and policies
GS41. importance of exercising restraint while expressing dissent and during conflict situations
GS42. how to avoid and manage distractions to be disciplined at work
GS43. importance of time management for achieving better results
GS44. work in a team in order to achieve better results
GS45. identify and clarify work roles within a team
GS46. communicate and cooperate with others in the team
GS47. seek assistance from fellow team members

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Working Safely at all times	**8**	**17**	**-**	**-**
PC1. work safely at all times, complying with health and safety legislation, regulations and other relevant guidelinessafety precautions (general) are: general workshop safety fire prevention general hazards manual lifting overhead lifting / mechanised lifting surface conditions stability of surrounding structures, furniture, etc.safety precautions (tig welding) are: protection from live and other electrical components, including insulation, proper earthing, proper loading, etc. proper handling and placement of hot metal adequate lighting appropriate personal protective equipment suitable aprons welding gloves safety boots correctly fitting overalls suitable welding helmet protection of self and others from the effects of the welding arc fume extraction/control measures safety measures for elevated and trench working reduction in the local air concentration due to release of argon gas during welding in confined places	2	4	-	-
PC2. take necessary safety precautions for tig welding operations	2	4	-	-
PC3. adhere to procedures and system in place for health and safety, pper and other regulations	2	4	-	-
PC4. check all connections of machines, welding leads, gas connection arrangement, electrode holder	2	5	-	-
Preparing for welding operations	**56**	**125**	**-**	**-**

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC5. interpret weld procedure data sheets specificationswelding concepts and mechanisms used are: rated output (duty cycle) measurement of electrical output and continuity relationship between wire feed speed control and welding current power source characteristics (volt/ampere graph, flat characteristic, constantvoltage output) types of current ac and dc and polarity ac welding (square wave forms and wave balancing) dc pulsed tig welding function of induction (principle, effect, fixed, stepped, variable control) return earth indirect control of welding current relay for electrical powerwelding techniques used are: fine adjustment of parameters (current and gas flow) selection of gas nozzle if required selection of the outer nozzle correct manipulation of the torch blending in stops/starts and tack welds starting techniques	5	10	-	-
PC6. select welding machines e.g. inverters, rectifiers and generators, according to the materials and task	5	5	-	-
PC7. select proper welding torch and electrode(w) that meet the job requirement and specification, select tungsten electrode by the colour of the tip according to base metal, and correct diameterselection and preparation of tungsten electrode are: types and classification of tungsten electrodes for different materials angle and technique of preparation of the tungsten electrode tips selection of the tungsten electrode diameter as per currenttorch components are: cables water cooled cables ceramic nozzle collet collet holder gas lens	5	10	-	-
PC8. obtain filler wire according to specifications	2	5	-	-
PC9. prepare for the tig welding process	-	5	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. prepare the materials and joint in readiness for weldingmaterial and joint preparation activities are: made rust free cleaned free from scaling, paint, oil/grease chemical cleaning made dry and free from moisture edges to be welded prepared as per job requirement (e.g. flat, square or bevelled) use various machines and techniques for the above (e.g. chamfering machine grinding and stripping, gas and plasma cutting, etc.) correctly positioned positioning: devices and techniques jigs and fixtures restraining devices such as clamps and weights/blocks setting up the joint in the correct position and alignment	2	5	-	-
PC11. select and fit the welding shielding gases for a range of given applicationsincluding back purgingshielding gases: shielding gases for gtaw applications for shielding gases/gas mixtures (argon, argon/helium mixtures,argon/hydrogen mixtures, nitrogen argon/nitrogen mixtures) gas pressure requirements flow rates for applications back purgingshielding gases equipment are: cylinders manifold systems regulators (fixed, single stage, two-stage) gas flow meters gas tubes and connectors use of solenoid valves economisers	5	5	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. plan the welding activities before they start them effectively and efficiently forachieving specifications as per wpsinterpreting the wps: welding process (iso codes for e.g. aws/asme) parent metal consumables pre welding joint preparation (cleaning, edge preparation, assembly, pre-heat) welding parameters welding positions (en iso 6947 pa, pb, pc, pd, pe, pf, pg; asme ix i-6 g/1-6 f) number and arrangement of runs to fully fill/weld joints electrode (w) filler wire electrical conditions required (type of current, alternating [a.c.] direct [d.c.], electrode polarity (negative), welding current ranges, electrode polarity (positive) methods of arc ignition (scratch, high frequency, lift start), carbon block shielding gas (type, flow rate, pre-weld gas flow, post-weld gas flow), techniques (including autogenous) control of heat input inter-pass/run cleaning/back gouging methods, post welding activities (wiring brushing, removal of excess weld metal where required), post-weld heat treatment (normalising, stress relief) where permissible and restrictions activities to be checked before start of welding are: correct set-up of the joint proper condition of electrical connections welding return and earthing arrangements operating parameters	2	5	-	-
PC13. connect torches and components	-	5	-	-
PC14. connect and adjust regulators and flow meters to cylinders	-	5	-	-
PC15. read, set and adjust current (amperage) as required	5	10	-	-
PC16. set pre-purge with shielding gas as required	5	10	-	-
PC17. prepare tungsten by sharpening or balling it to desired tip shape	-	5	-	-
PC18. set and verify gas flow rates	5	10	-	-
PC19. prepare and support the joint, using the appropriate methods	5	10	-	-
PC20. tack weld the joint at appropriate intervals, and check the joint for accuracy before final welding, wherever required	5	10	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC21. match feed and travel speed as required	5	10	-	-
Carrying out welding operations	**40**	**80**	**-**	**-**
PC22. perform tig welding operations to meet welding procedure specification requirementsbasic principles of tig welding are: the arc burns between a non-consumable tungsten electrode and the work piece exclusively inert gases (argon, helium) are used as shielding gases and other gases and gas mixtures tig welding installation for most applications an electrode with a negative polarity is used for welding of aluminium, alternating current must be used for arc ignition a high-frequency high voltage is used	5	10	-	-
PC23. use correct technique for starting the arc (using hf (high frequency) unit, scratching the electrode on the job material, lifting the electrode immediately after touching the job material)	5	10	-	-
PC24. use correct angle of torch and filler wire, direction of weld and inclusion defect	5	10	-	-
PC25. weld the joint to the specified quality, dimensions and profile.	5	10	-	-
PC26. use manual welding and related equipment, to carry out tig welding processes.welding equipment are: rectifier (pulsing) inverter generator equipment for current regulation high frequency unit torches electrodes filler wires water cooling and circulation system for tig torch (water/air cooled torch) return clamps foot pedal ancillary equipment (tungsten tip grinder for tungsten electrode, wirebrushes, linishers, hammer, power saw, angle, pedestal and straightgrinders, chisel) other equipments such as holding, jig fixtures, measuring equipments etc.	5	10	-	-
PC27. use welding consumables appropriate to the material and application, to include ac current types and dc current types	5	10	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC28. produce joints of the required quality and of specified dimensional accuracy which achieve a weld quality equivalent to level b of iso 5817weld quality check standards are: required parameters for dimensional accuracy weld finishes are built up to the full section of the weld joins at stop/start positions merge smoothly weld surface is o free from cracks substantially free from porosity free from any pronounced hump or crater substantially free from shrinkage cavities substantially free from arcing or chipping marks fillet welds are o equal in leg length slightly convex in profile (where applicable and preferable) size of the fillet equivalent to the lower thickness of the material welded weld contour is of linear and of uniform profile smooth and free from excessive undulations regular and has an even ripple formation welds are adequately fused, and there is minimal undercut, overlap and surface inclusions tack welds are blended in to form part of the finished weld, without excessive hump corner joints have minimal burn through to the underside of the joint or, where appropriatetypes of joints are: fillet lap joints tee fillet joints corner joints butt joints square single vee double vee	5	10	-	-
PC29. roduce joints from various materials in different forms.materials used for welding are: carbon steel stainless steel (all grades) aluminium and aluminium alloys nickel and nickel alloys titanium copper and copper alloys forms on which welding done are: sheet (less than 1.5 mm) plate (8 mm) section pipe/tube other forms	5	10	-	-
Testing for quality	**19**	**40**	**-**	**-**
PC30. use appropriate methods and equipment to check the quality, and that all dimensional and geometrical aspects of the weld are to the specification	5	10	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC31. check that the welded joint conforms to the specification, by checking various quality parameters using visual inspectionquality parameters are: shape and length of the draglines smoothness of the sides sharpness of the top edges amount of slag adhering to the metalquality parameters a to be checked are: dimensional accuracy alignment/squareness size and profile of weld visual defects ndt/dt tested defects types of visual inspections are: use of visual techniques lighting low powered magnification fillet weld gauges	5	10	-	-
PC32. identify various weld defects;types of weld defects are: lack of continuity of the weld uneven and irregular ripple formation incorrect weld size or profile undercutting overlap inclusions (tungsten) porosity internal cracks surface cracks lack of fusion lack of penetration gouges stray arc strikes sharp edgeswelding consumables used are: filler wires for different base materials shielding gasconsumables classification as per: sizes [diameters, lengths] strength and elongation of the weld metal impact properties of the weld metal chemical composition of the weld metal protection of bare wires	2	5	-	-
PC33. detect surface imperfections and deal with them appropriately	2	5	-	-
PC34. carry out dpt tests to assess fine defect open to the surface not detected by visual inspection (vt)	5	10	-	-
Post welding techniques	**15**	**30**	**-**	**-**
PC35. prepare for non-destructive testing of the welds for a range of testsnon-destructive tests (ndt) are: visual inspection leak test dye penetrant (dpt) fluorescent penetrant (fpt) magnetic particle (mpt) radiographic (rt) ultrasonic (ut)	5	10	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC36. prepare for destructive tests on weld specimens for select testsdestructive tests (dt) are: nick break test bend tests (such as face, root or side, as appropriate) metallographic (micro structure, haz, etc.) mechanical (peel, tensile and shear, fatigue, impact tests) and hardness in different zones chemicalhandling specimens for tests: handling hot materials using chemicals for cleaning and etching using equipment to fracture welds	5	10	-	-
PC37. shut down and make safe the welding equipment and gases on completion of the welding activities, clean the area & store the accessories in designated place	5	10	-	-
Dealing with Contingencies	**5**	**15**	**-**	**-**
PC38. detect equipment malfunctions and deal with them appropriately	5	10	-	-
PC39. deal promptly and effectively with problems within their control, and seek help and guidance from the relevant people if they have problems that they cannot resolve	-	5	-	-
NOS Total	**143**	**307**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	ISC/N0911
NOS Name	Perform Tungsten Inert Gas (TIG) Welding also known as Gas Tungsten Arc Welding (GTAW)
Sector	Iron and Steel
Sub-Sector	Steel, Sponge Iron, Ferro Alloys, Re-Rollers, Refractory
Occupation	Mechanical Maintenance
NSQF Level	4
Credits	TBD
Version	1.0
Last Reviewed Date	30/12/2014
Next Review Date	31/03/2022
NSQC Clearance Date	18/06/2015

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.

2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.

3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.

4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).

5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.

6. To pass the Qualification Pack assessment, every trainee should score the Recommended Pass % aggregate for the QP.

7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

Minimum Aggregate Passing % at QP Level : 60

(**Please note**: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
ISC/N0008.Use basic health and safety practices at the workplace	45	105	-	-	150	15
ISC/N0009.Work effectively with others	30	70	-	-	100	10
ISC/N0910.Manually cut metal and metal alloys using oxy-fuel gases	85	215	-	-	300	30
ISC/N0911.Perform Tungsten Inert Gas (TIG) Welding also known as Gas Tungsten Arc Welding (GTAW)	143	307	-	-	450	45
Total	**303**	**697**	**-**	**-**	**1000**	**100**

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
5S	Technique of maintaining orderliness â€“Japanese terminology.
CP	Control Plan.
WI	Work Instructions.

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.